

METHODS OF QUANTUM FIELD THEORY FOR T=0

NISHTHA TIKALAL

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1 The Interaction Representation

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1. THE INTERACTION REPRESENTATION

We seek an intuitive & simplistic method for writing down any term in a perturbation series. The most convenient method is the diagram technique (borrowed from QFT).

We begin by introducing “particle-field” operators:

$$\begin{aligned}\psi(\xi) &= \sum_i \varphi_i(\xi) a_i \\ \psi^+(\xi) &= \sum_i \varphi_i^* a_i^+\end{aligned}\tag{1}$$

From chpt. 1, we previously defined a_i & a_i^+ as the creation/annihilation operators. $\varphi_i(\xi)$ is defined as the wavefunction of a particle in state i . Similarly the operators in (1) obey the following commutation relations:

$$\begin{aligned}\psi(\xi)\psi^+(\xi') \mp \psi^+(\xi')\psi(\xi) &= \delta(\xi - \xi') \\ \psi(\xi)\psi(\xi') \mp \psi(\xi')\psi(\xi) &= 0 \\ \psi^+(\xi)\psi^+(\xi') \mp \psi^+(\xi')\psi^+(\xi) &= 0\end{aligned}\tag{2}$$

The upper sign of $\mp$ refers to Bose statistics & the lower sign Fermi statistics. From this new representation, a one-particle operator $F^{(1)}$ is of the form:

$$F^{(1)} = \int \psi^+(\xi) f^{(1)} \psi(\xi) d\xi\tag{3}$$

In terms of the operators ψ & ψ^+ , the Hamiltonian can be expressed as:

$$\begin{aligned}H &= \int \left[\frac{1}{2m} \nabla \psi_\alpha^+(\vec{r}) \cdot \nabla \psi_\alpha(\vec{r}) + U(\vec{r}) \psi_\alpha^+(\vec{r}) \psi_\alpha(\vec{r}) \right] d\vec{r} \\ &+ \frac{1}{2} \int \psi_\alpha^+(\vec{r}) \psi_\beta^+(\vec{r}') U^{(2)}(\vec{r}, \vec{r}') \psi_\beta(\vec{r}') \psi_\alpha(\vec{r}) d\vec{r} d\vec{r}' + \dots\end{aligned}\tag{4}$$

For a system of particles with 1/2 spin in the absence of a magnetic field. From (4), the following assumptions are made:

i) interaction between particles are spin-independent.

ii) α & β represent spin projection, & the expression is summed over pairs of identical indices.

(4) takes the same expression for the average energy of N particles in identical states $\psi_\alpha(\vec{r})$, normalized by the relation:

$$\int |\psi_\alpha(\vec{r})|^2 d\vec{r} = N$$

This property allows us to easily identify the Hamiltonian in the second quantization. Another operator of importance is:

$$n(\vec{r}) = \sum_a \delta(\vec{r} - \vec{r}_a),$$

which is the operator for the density of particles at a given point. By its form, we can apply (2) to obtain:

$$\begin{aligned} n(\vec{r}) &= \int \psi_\alpha^\dagger(\vec{r}_a) \delta(\vec{r} - \vec{r}_a) \psi_\alpha(\vec{r}_a) d\vec{r}_a \\ &= \psi_\alpha^\dagger(\vec{r}) \psi_\alpha \end{aligned} \quad (5)$$

Familiarly, the operator for the total number of particles has the form:

$$N = \int n(\vec{r}) d\vec{r} = \int \psi_\alpha^\dagger(\vec{r}) \psi_\alpha(\vec{r}) d\vec{r}$$

Suppose we have a system of particles with a Hamiltonian H . We want to determine how the state of the system changes with time. $\therefore$ we must solve the Schrödinger equation:

$$i \frac{\partial \Phi}{\partial t} = H \Phi \quad (6)$$

recall that the solution takes the form:

$$\Phi(t) = e^{-iHt} \Phi_H \quad (7)$$

where Φ_H is a time independent function. To determine the matrix elements of the operator F vary with time,

$$\begin{aligned} F_{nm}(t) &= \langle \Phi_n^*(t) F \Phi_m(t) \rangle \\ &= \langle \Phi_{Hn}^* e^{iHt} F e^{-iHt} \Phi_{Hm} \rangle \end{aligned} \quad (8)$$

(8) can be interpreted as the matrix element of the operator

$$\tilde{F}(t) = e^{iHt} F e^{-iHt} \quad (9)$$

with respect to Φ_H . This corresponds to a new representation called the Heisenberg representation.

The most important property of this representation is that the wavefunctions Φ_H are time-independent. Time-dependence is transferred to the operators. By (9),

$$\frac{\partial \tilde{F}}{\partial t} = i(H\tilde{F} - \tilde{F}H) \equiv i[H, \tilde{F}] \quad (10)$$

For stationary states, the wavefunction Φ_{Hn} satisfies the equation:

$$H\Phi_{Hn} = E_n\Phi_{Hn} \quad (11)$$

By (8), we have:

$$F_{mn}(t) = \langle \Phi_{Hn}^* F \Phi_{Hm} \rangle e^{i(E_n - E_m)t} \quad (12)$$

For a wavefunction in a state i in a system of non-interacting particles, let:

$$\varphi_i(\xi) = \frac{1}{\sqrt{V}} e^{i\vec{p} \cdot \vec{r}}$$

Here, V is the volume. The operator Ψ takes the form:

$$\Psi = \frac{1}{\sqrt{V}} \sum_p a_p e^{i\vec{p} \cdot \vec{r}} \quad (13)$$

By (4), the Hamiltonian reads:

$$H = \sum_p \varepsilon_0(p) n_p;$$

recall $\varepsilon_0(p)$ is the free particle energy. By (9), the operator $\tilde{\Psi}(\vec{r}, t)$ is:

$$\begin{aligned} \tilde{\Psi}(\vec{r}, t) &= \frac{1}{\sqrt{V}} \sum_p e^{i \sum_{p'} \varepsilon_0(p') n_{p'} t} a_p e^{-i \sum_{p''} \varepsilon_0(p'') n_{p''} t} e^{i\vec{p} \cdot \vec{r}} \\ &= \frac{1}{\sqrt{V}} \sum_p a_p e^{i[\vec{p} \cdot \vec{r} - \varepsilon_0(p)t]} \end{aligned} \quad (14)$$

(14) is the Heisenberg representation. Note that it does not satisfy the commutation rules as in the Schrödinger representation (2).

Beyond the Heisenberg & Schrödinger representation, there is a 3rd one known as the interaction representation. We begin by isolating parts of the Hamiltonian:

$$H = H_0 + H_{\text{int}} \quad (15)$$

where the subscript int denotes “interaction between the particles.” We then utilize the following transformation of the Schrödinger wave function:

$$\Phi_i = e^{iH_0 t} \Phi \quad (16)$$

Taking $\frac{\partial \Phi_i}{\partial t}$, we obtain:

$$\begin{aligned} \frac{\partial \Phi_i}{\partial t} &= \frac{\partial}{\partial t} [e^{iH_0 t} \Phi] \\ &= \frac{\partial}{\partial t} [e^{iH_0 t}] \Phi + e^{iH_0 t} \frac{\partial}{\partial t} [\Phi] \\ &= iH_0 e^{iH_0 t} \Phi + e^{iH_0 t} \frac{\partial \Phi}{\partial t} \end{aligned}$$

Multiplying both sides by i ,

$$\begin{aligned} i \frac{\partial \Phi_i}{\partial t} &= i \left(iH_0 e^{iH_0 t} \Phi \right) + i e^{iH_0 t} \frac{\partial \Phi}{\partial t} \\ &= -H_0 e^{iH_0 t} \Phi + e^{iH_0 t} i \frac{\partial \Phi}{\partial t} \\ &= -H_0 e^{iH_0 t} \Phi + e^{iH_0 t} (H_0 + H_{\text{int}}) \Phi \\ &= -H_0 e^{iH_0 t} \Phi + e^{iH_0 t} H_0 \Phi + e^{iH_0 t} H_{\text{int}} \Phi \\ &= \underbrace{-H_0 e^{iH_0 t} \Phi + H_0 e^{iH_0 t} \Phi}_0 + e^{iH_0 t} H_{\text{int}} \Phi \\ &= e^{iH_0 t} H_{\text{int}} \Phi = H_{\text{int}} e^{iH_0 t} \Phi \end{aligned} \quad (17)$$

We can further re-write this expression as:

$$i \frac{\partial \Phi_i}{\partial t} = H_{\text{int}}(t) \Phi_i \quad (18)$$

$$H_{\text{int}}(t) = e^{iH_0 t} H_{\text{int}} e^{-iH_0 t}$$

The operator $F(t)$ satisfies:

$$\frac{\partial F(t)}{\partial t} = i[H_0, F(t)] \quad (19)$$

We now explore how $\Phi_i(t)$ is time dependent in the interaction representation. Since the operators of $H_{\text{int}}(t)$ do not commute, we can't simply write

$$\Phi_i(t) = C e^{-i \int_{t_0}^t H_{\text{int}}(t') dt'}$$

as a solution for (17), where C is a constant. Rather, we begin with (17):

$$i \frac{\partial \Phi_i}{\partial t} = H_{\text{int}}(t) \Phi_i$$

By integrating both sides between the limits of t_0 to t ($t > t_0$),

$$\int_{t_0}^t i \frac{\partial \Phi_i}{\partial t} = \int_{t_0}^t H_{\text{int}}(t') \Phi_i(t') dt'$$

$$\Phi_i(t) = \Phi_i(t_0) - i \int_{t_0}^t H_{\text{int}}(t') \Phi_i(t') dt'$$

The solution can be written as a series:

$$\Phi_i(t) = \Phi_i^{(0)}(t) + \Phi_i^{(1)}(t) + \dots$$

in H_{int} with $\Phi_i^{(0)}(t) = \Phi_i^{(0)}(t_0)$, which is the zeroth approximation. The first approximation is:

$$\Phi_i^{(1)}(t) = -i \int_{t_0}^t H_{\text{int}}(t_1) dt_1 \Phi_i(t_0)$$

The second approximation is:

$$\Phi_i^{(2)}(t) = - \int_{t_0}^t H_{\text{int}}(t_1) dt_1 \int_{t_0}^{t_1} H_{\text{int}}(t_2) dt_2 \Phi_i(t_0).$$

The n^{th} approximation:

$$\Phi_i^{(n)}(t) = (-i)^n \int_{t_0}^t H_{\text{int}}(t_1) dt_1 \int_{t_0}^{t_1} H_{\text{int}}(t_2) dt_2 \dots \int_{t_0}^{t_{n-1}} H_{\text{int}}(t_n) dt_n \Phi_i(t_0)$$

The whole result can be represented in the form

$$\Phi_i(t) = S(t, t_0) \Phi_i(t_0) \quad (20)$$

$S(t, t_0)$ is a matrix given by the series:

$$S(t, t_0) = 1 - i \int_{t_0}^t H_{\text{int}}(t_1) dt_1 + \dots + (-i)^n \int_{t_0}^t H_{\text{int}}(t_1) dt_1 \dots \int_{t_0}^{t_{n-1}} H_{\text{int}}(t_n) dt_n + \dots \quad (21)$$

To make (21) more symmetric, we consider the n^{th} term of the series:

$$(-i)^n \int_{t > t_1 > \dots > t_0} H_{\text{int}}(t_1) H_{\text{int}}(t_2) \dots H_{\text{int}}(t_n) dt_1 dt_2 \dots dt_n$$

We define T as the operation which achieves the following: carrying out all permutation of the variables t_n , adding the resulting expression & dividing by $n!$ $\therefore$ (21) becomes

$$\begin{aligned} S^{(n)}(t, t_0) &= \frac{(-i)^n}{n!} \int_{t_0}^t \dots \int_{t_0}^t T(H_{\text{int}}(t_1) \dots H_{\text{int}}(t_n)) dt_1 \dots dt_n \quad , \quad \& \\ S(t, t_0) &= T e^{-i \int_{t_0}^t H_{\text{int}}(t') dt'} \dots \end{aligned} \quad (22)$$

$S(t, t_0)$ has the property:

$$S(t_2, t_1) S(t_1, t_0) = S(t_2, t_0), \quad t_2 > t_1 > t_0 \dots \quad (23)$$

To summarize, (16) & (18) establish the relationship between the Schrödinger equation & the interaction representation. (20) establishes the relationship between the interaction representation & the Heisenberg representation.

Suppose we have a wavefunction that transforms as described:

$$\Phi_i(t) = Q(t) \Phi_H \dots \quad (24)$$

where $Q(t)$ is a unitary operator. By (20),

$$Q(t) = S(t, t_0) Q(t_0) \quad , \quad ; \quad \text{by (24),}$$

$$Q(t) = S(t, \alpha) P.$$

Here, α is a constant of time, & P is a time-independent operator. To find P , we substitute (24) into eqs (16) & (7). Recall the three formulas below:

$$\begin{cases} (24): & \Phi_i(t) = Q(t) \Phi_H \\ (16): & \Phi_i = e^{iH_0 t} \Phi \\ (7): & \Phi(t) = e^{-iHt} \Phi_H \end{cases}$$

$$e^{iH_0 t} \Phi = Q(t) \Phi_H$$

$$e^{iH_0 t} e^{-iHt} \Phi_H = Q(t) \Phi_H$$

Substituting $Q(t)$:

$$e^{iH_0 t} e^{-iHt} \Phi_H = S(t, \alpha) P \Phi_H$$

$$e^{iH_0 t} = S(t, \alpha) P e^{iHt}$$

Since $S(\alpha, \alpha) = 1$,

$$e^{iH_0 \alpha} e^{-iH \alpha} = P$$

We now introduce the concept of an “adiabatically turned on interaction.” Suppose @ time $t = -\infty$, particle interaction does not occur. After “turning on” the interaction, $P \rightarrow 1$ as $\alpha \rightarrow -\infty$. Thus,

$$\Phi_i(t) = S(t) \Phi_H \quad (25)$$

where

$$S(t) = S(t, -\infty) \quad (26)$$

By the property of $S(t)$ in (23),

$$S(t_2, t_1) = S(t_2)S^{-1}(t_1) \quad (27)$$

The relation between the operators in the interaction representation & Heisenberg operators takes the form:

$$\tilde{F}(t) = S^{-1}(t)F(t)S(t) \quad (28)$$

We note the products of several Heisenberg operators, averaged over the ground state Φ_H :

$$\langle \Phi_H^* T \left(\tilde{A}(t) \tilde{B}(t') \tilde{C}(t'') \dots \right) \Phi_H \rangle \quad (29)$$

To generalize the definition of T-ordering in Fermi operators, we take the product of $A(t_1), B(t_2), \dots, C(t_n)$, multiplied by a factor of $(-1)^P$, where P is the number of permutations. Letting $F(t_i)$ denote Fermi operators & $B(t_i)$ denote the Bose operators. Thus,

$$\begin{aligned} T(F_1(t_1)F_2(t_2)) &= \begin{cases} F_1(t_1)F_2(t_2), & t_1 > t_2 \\ -F_2(t_2)F_1(t_1), & t_1 < t_2 \end{cases} \\ T(B_1(t_3)F_1(t_1)) &= \begin{cases} B_1(t_3)F_1(t_1), & t_3 > t_1 \\ F_1(t_1)B_1(t_3), & t_3 < t_1 \end{cases} \\ T(B_2(t_4)B_1(t_3)) &= \begin{cases} B_2(t_4)B_1(t_3), & t_3 < t_4 \\ B_1(t_3)B_2(t_4), & t_3 > t_4 \end{cases} \end{aligned}$$

Suppose the arrangement of times in (29) is: $t > t' > t''$. By (28),

$$\begin{aligned} &\langle \Phi_H^{\circ*} S^{-1}(t) A(t) S(t) S^{-1}(t') B(t') S(t') \dots \Phi_H^{\circ} \rangle \\ &= \langle \Phi_H^{\circ*} S^{-1}(\infty) S(\infty, t) A(t) S(t, t') B(t') \dots \Phi_H^{\circ} \rangle \\ &= \langle \Phi_H^{\circ*} S^{-1}(\infty) T(A(t) B(t') C(t'') \dots S(\infty)) \Phi_H^{\circ} \rangle \end{aligned} \quad (30)$$

From (30), the only term left to be determined is:

$$\Phi_H^{\circ*} S^{-1}(\infty) = [S(\infty) \Phi_H^{\circ}]^*$$

By (20) & (25),

$$\Phi_H^{\circ} = \Phi_i(-\infty) \quad ; \quad S(\infty) \Phi_H^{\circ} = \Phi(\infty)$$

Recall from Quantum mechanics that a system in a non-degenerate stationary states cannot transition to another state under a slow perturbation. Thus, for the function $\Phi_i(\infty) = S(\infty) \Phi_H^{\circ}$, it can only differ from Φ_H° by the phase factor

$$S(\infty) \Phi_H^{\circ} = e^{iL} \Phi_H^{\circ} \quad (31)$$

The final result is thus:

$$\begin{aligned} &\langle \Phi_H^{\circ*} T \left(\tilde{A}(t) \tilde{B}(t') \tilde{C}(t'') \dots \right) \Phi_H^{\circ} \rangle \\ &= \frac{\langle \Phi_H^{\circ*} T \left(\tilde{A}(t) \tilde{B}(t') \tilde{C}(t'') \dots S(\infty) \right) \Phi_H^{\circ} \rangle}{\langle \Phi_H^{\circ*} S(\infty) \Phi_H^{\circ} \rangle} \end{aligned} \quad (32)$$

REFERENCES

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